Study 3 results: visual clutter and counting

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Overview

We tested whether visual clutter makes a vision-language model (Qwen3-VL-4B) worse at counting shapes and objects and at following lines, and whether clutter the model can put a name to hurts more than clutter it can't. The run worked as planned: all 12,500 images across Mohid's four datasets got a readable answer, and our accuracy numbers match each dataset's official scorer exactly. The clearest findings are that every kind of clutter hurts (H3), clutter that looks like the real thing hurts most (accuracy falls from 97% to 31% on Objects and from 45% to 3% on Shapes), both nameability and visual similarity matter (H1), the model counts a broken shape as a whole one about half the time (H5), it reads and copies written words in the picture (H6), and colouring the lines helps it follow them (H8). The caveats are real, though: even with no clutter the 4B model gets only 42–45% of shape images right, so several cluttered versions sit near 0% and are hard to tell apart; H2's nameability test was weak because the model gives a name to almost every squiggle; H4 and H6 hold in one dataset but not the other; and on Lines the model mostly assumes a line runs straight across instead of tracing it, so H7's crossings test did not measure what we meant it to.

Setup

One model, four datasets, every image, fixed in the plan (docs/study3_methodology.pdf) before the run.

- **Model:** Qwen3-VL-4B-Instruct (pinned revision ebb281ec), 16-bit, one Kaggle T4 GPU, greedy decoding. The plan started with the 2B model, but it got only 15% of clean Shapes images right, so the plan's floor rule moved every dataset to 4B.
- **Data:** Mohid's four datasets from the atml-shapes-vqa organisation: Shapes (4 versions), Shapes Hard (10 versions, incl. other_shapes), Objects (5) and Lines (6). All 500 scenes of every version, so 12,500 images.
- **Answer start:** each reply is started with the first word of the answer format ("circle:", "apple:", "box:") so the model answers instead of explaining. Every version of a dataset gets the same start, so versions stay comparable.

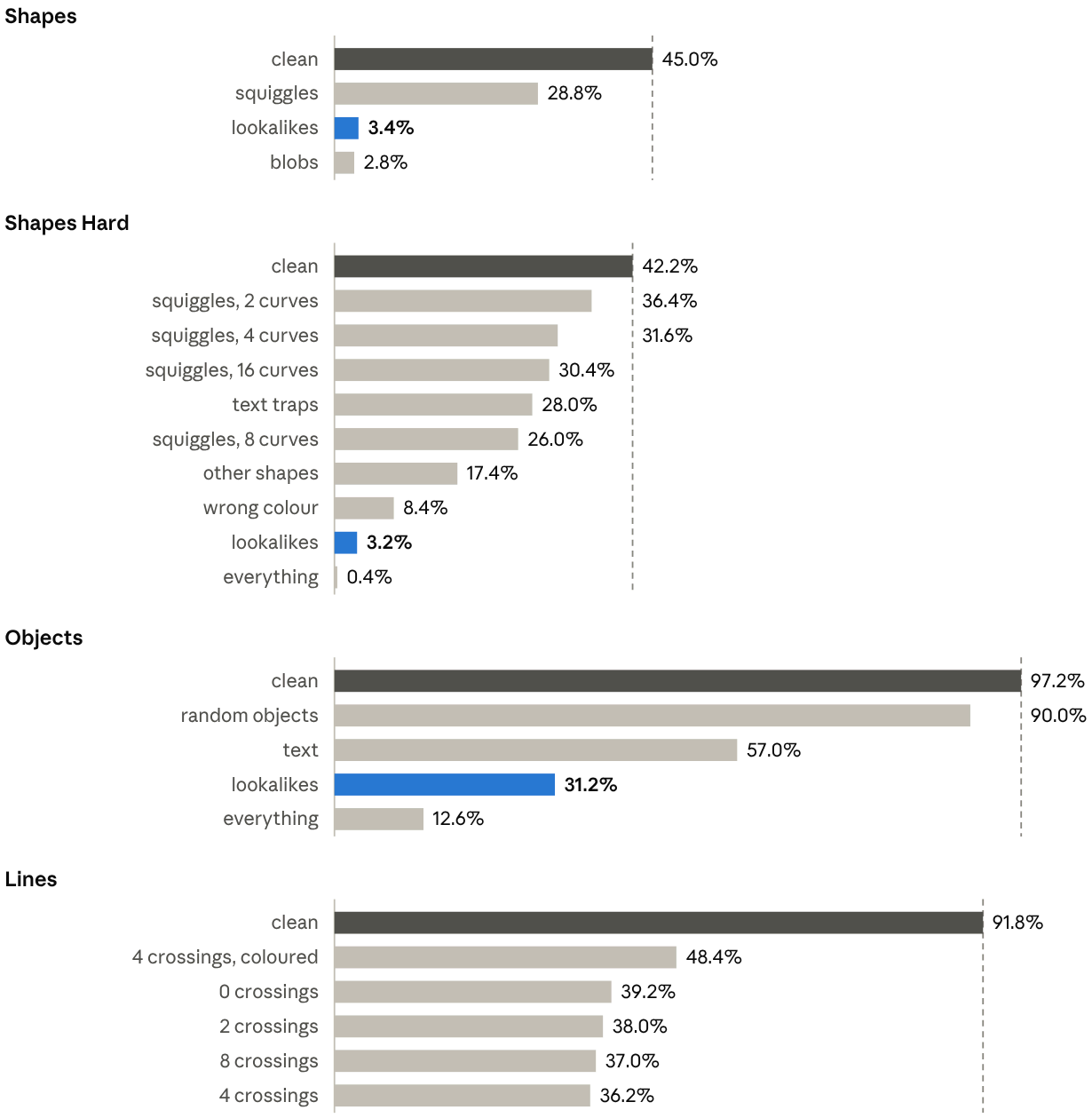
- **Scoring:** an image counts as right only if every count (or the line's end box) is right, scored by each dataset's own vqa_scoring.py.
- **Uncertainty:** each result has a 95% range from 10,000 bootstrap resamples over scenes. A result is "above 0" only if its whole range is above 0.
- **Cost:** about 4.9 GPU hours for the predictions; the analysis runs on CPU.

Accuracy by dataset and version

Lookalikes are the worst clutter in every dataset that has them, and clean accuracy itself is low for the two shape datasets.

Lookalike clutter does the most damage: Objects falls from 97% to 31%

Share of images with every count right (%), Qwen3-VL-4B, 500 scenes per version. Blue = lookalikes; dashed line = clean.



Kaggle notebook clutter-vlm study3 main, scored by each dataset's vqa_scoring.py · 12,500 images

Objects and Lines start high (97% and 92% clean), so drops there are easy to see. Shapes and Shapes Hard start at 45% and 42%, which is the floor problem described under Caveats.

Results by hypothesis

Three hypotheses are supported outright, three are mixed or partly supported, and one came out the opposite way. H2 has its own section below. Numbers are accuracy points (percentage points) unless the row says otherwise; brackets are the 95% range.

Hypothesis	What we predicted	What we found	Verdict
H1 Names vs looks	Clutter the model can name hurts more than clutter it can't	Named shapes (heart, moon, cross, bolt) vs squiggles: 13.7 more points lost [10.2, 17.2]. Lookalikes vs named shapes: 14.2 more [10.8, 17.8]. Objects lookalikes vs random objects: 58.8 more [54.2, 63.2]	Supported: both names and looks matter
H3 Every clutter hurts	Every cluttered version is below clean	All 21 drops above 0; smallest is 2-curve squiggles, 5.8 points, with its range starting at 1.4	Supported
H4 Lookalikes overcount most	Lookalikes add the most extra counts	Shapes: 0.46 more extra counts per image than blobs [0.34, 0.58]. Shapes Hard vs text: +0.19 [−0.01, 0.39]. Objects vs text: −0.02 [−0.23, 0.17]	Mixed: Shapes only
H5 Broken shapes counted as whole	Incomplete shapes are counted as complete shapes easily (rate above 50% and above other clutter)	Counted as whole 52% of the time on Shapes [49, 54] and 50% on Shapes Hard [47, 52]; 23–32 points above the rate for other clutter	Partly: far above other clutter, not clearly above 50%

Hypothesis	What we predicted	What we found	Verdict
H6 Text	Written words hurt more than the same amount of non-text clutter, and the model copies them	Shapes Hard text vs squiggles: +3.1 [−1.5, 7.5]. Objects text vs random objects: +33.0 [28.2, 37.8]. A word naming a shape raises that shape's count in 23% of cases vs 2% without it (Shapes Hard), 20% vs 0.4% (Objects)	Mixed on harm; copying supported
H7 Line crossings	More crossings hurt more when the line ends in a different position	+1.4 points per crossing [0.3, 2.4]	Opposite of prediction
H8 Colour	Coloured lines are easier to follow	+12.2 points vs plain lines with 4 crossings [8.8, 15.8]	Supported

H5 by shape. How often a broken shape is counted as whole depends heavily on the shape: stars 92%, triangles 47%, circles 36%, squares 25% (Shapes; Shapes Hard is within 6 points of each). Shapes broken by a gap fool the model more often (55–57%) than shapes with a wavy edge (45–47%).

Why H7 came out backwards. On images where the line ends straight across from where it starts, the model is right about half the time (46–51%). Where the line ends somewhere else, it is right only 7–17% of the time. So it mostly guesses "straight across" instead of tracing the line, and extra crossings can't make a guess much worse. With coloured lines its accuracy on these harder images rises to 30%, which is where H8's gain comes from.

The H2 situation, in simple terms

H2's main test gave a clear answer (busier squiggles hurt more), but the test meant to explain why didn't work, because the model finds a name for almost every squiggle.

The question. Shapes Hard has squiggles with 2, 4, 8 and 16 curves. We had two competing ideas. "Busy clutter": more curves make the picture busier, so accuracy should go down. "Names": a more complicated squiggle is harder to name, so the model should ignore it and accuracy should stay the same or go up.

What happened. Accuracy was 36% with 2 curves, 32% with 4, 26% with 8 and 30% with 16. On average it falls 2.4 points each time the curves double [−3.8, −0.8]. By the rule we fixed in the plan, that means the "busy clutter" idea wins. (The small rise at 16 curves is within the noise; we don't read anything into it.)

Why we can't fully separate the two ideas. The "names" idea assumes complicated squiggles are harder to name. To check that instead of assuming it, we cut each squiggle out of its picture and asked the model five times what it looks like. Its nameability score is the share of answers that name a real thing rather than "squiggle", "curve" and so on. The model almost always found a name: "C", "S", "U", "Arrow". Average scores were 0.77, 0.79, 0.74 and 0.97 for 2, 4, 8 and 16 curves. So nameability barely changed between versions, and the 16-curve squiggles were actually the easiest to name.

What the untangling test says. We then let curves and nameability compete to explain accuracy. Holding nameability fixed, each doubling of curves still costs 2.7 points [−4.3, −1.1]. Holding curves fixed, nameability's effect is +6.7 points with a range from −4.3 to +17.8, so we can't tell whether it matters at all.

Bottom line. Busyness clearly matters. Names might also matter, but these squiggles never became hard enough to name for us to see it. The sanity check passed: the model names hearts, moons, crosses and lightning bolts correctly every time (scores 0.999). A sharper test would need squiggles the model genuinely can't name, or human nameability ratings, rather than counting letters like "C" and "S" as names.

Caveats and fixes

The biggest limit is how weak the model is at counting shapes even without clutter; the rest are smaller.

- **Floor effect.** The model gets only 45% (Shapes) and 42% (Shapes Hard) of clean images right. Lookalikes, blobs and "everything" all land at 0–3%, so we can't rank those against each other. Objects (97% clean) is the fairest test of our ideas.
- **One model.** Every result is for Qwen3-VL-4B only. We don't yet know whether a bigger or different model behaves the same.

- **Nameability is the model's own view.** The H2 score measures what this model can name, which is what we wanted, but it is not a human rating.
- **Answer start.** Starting each reply with "circle:" (or "apple:", "box:") makes answers readable. It might nudge the model, but it is the same for every version, so comparisons between versions stay fair.
- **Lines.** The model mostly guesses rather than traces (see H7), so the Lines results say more about guessing than about how clutter disrupts tracing.
- **Fixes made after the run (no result changed).** The analysis first crashed because count columns were read as text; we converted them to numbers and re-ran on CPU. The naming check first showed the lightning bolt as never named; the model actually answers "Lightning" every time, and our matching code was looking under the wrong label. After the fix all four new shapes pass. We also relabelled the H2 lines in the summary file to name the idea that wins.

Files

- **Code and plan:** github.com/zoyakhan51/clutter-vlm. The plan is docs/study3_methodology.pdf; results come from analysis code at commit 7a0d9dc (the later commit 72c46b6 only renames the H2 lines in the summary file).
- **Predictions:** Kaggle notebook "clutter-vlm study3 main", output folder preds (answers, scorer output, nameability answers).
- **Results:** Kaggle notebook "clutter-vlm study3 analysis", version 2, output folder results (results_qwen3vl4b.json with every number and range, and summary_qwen3vl4b.txt). Both notebooks are private to Zoya's Kaggle account for now.